

Software Requirements Specification (SRS) Prompts

For this chat, assume I'm a software engineering student wanting your help in creating good SE artifacts such as Requirements Specifications, Design Specs, Test Scenarios, etc.

I have these requirements provided to me for a POS system by my instructor. I have to iteratively refine it until they become numbered FR, BR, and NFR. First, please read this and ask me questions about parts of this POS system that may need more clarification before you can draft the requirements lists.

1. **Serve customers at the counter**
 - Cashiers can start a checkout, add or remove items, apply valid discounts, and see the running total.
 - A sale can be **cancelled** any time before it is finished.
2. **Accept payments**
 - Customers can pay **by cash** or **by credit card** which is processed via a **Payment Service Provider** (eg., K-Bank)
 - The system **works with the card machine** from, say, K-Bank to send the amount and receive approval
 - If a card is **declined**, the cashier can retry, switch to cash, or cancel the sale.
3. **Print documents: Receipt** for successful sale**, Refund Receipt** for refunds. Prints to store receipt printer.
4. Updates stock in **store's inventory service** app**: reduces** stock of the sold item, adds back stock of returned items.
5. **Handle returns and refunds. **
 - Cashiers or managers can find the original sale and **process a return**.
 - The system **refunds** the customer (cash or card, following store policy) and records the reason.
6. **Manager Functions**
 - Managers can view a **daily sales summary** for a selected date, including total cash & card amount received.
 - Managers can **add or update products and prices** to be used by POS.

Before drafting numbered Functional Requirements (FR), Business Rules (BR), and Non-Functional Requirements (NFR), I need clarification on the following points. Short answers are sufficient, and you may write "use your recommended assumption" where appropriate.

A. Users and access

1. Must cashiers and managers log in individually, or can the POS use a shared account?
2. Can managers perform all cashier functions?
3. Should important actions such as discounts, sale cancellations, refunds, and price changes be recorded with the employee, date, and time?

B. Checkout and products

4. How are products added to a sale: barcode scanning, product search, manual product-code entry, or all three?
5. Can the cashier change an item's quantity after adding it?
6. What should happen when there is insufficient stock?
 - Block the sale
 - Display a warning but allow it

- Rely entirely on the Inventory Service
- 7. Are taxes included in the product price, or must the POS calculate and add tax?
- 8. When is a sale considered finished: after successful payment, after receipt printing, or at another point?
- 9. When a sale is cancelled before completion, should the system record the cancelled sale and its reason, or simply discard it?

C. Discounts

- 10. What kinds of discounts are supported?
 - Percentage discount
 - Fixed-amount discount
 - Product-level discount
 - Whole-sale discount
- 11. Who may apply discounts: cashiers and managers, or managers only?
- 12. What makes a discount “valid”? For example, must it be active, within a valid date range, applicable to selected products, and below a maximum amount?
- 13. May multiple discounts be combined in one sale?

D. Payments

- 14. Can a customer split payment between cash and card, or must each sale use exactly one payment method?
- 15. For cash payment, should the cashier enter the amount received and have the POS calculate the change?
- 16. If communication with the card machine or Payment Service Provider times out, what should happen? This differs from a definite card decline because the payment result may be unknown.
- 17. After card approval, what should happen if completing the sale, updating inventory, or printing the receipt fails?
- 18. Is card settlement and end-of-day reconciliation within the POS scope, or does the Payment Service Provider handle it separately?

E. Receipts

- 19. What information must appear on receipts and refund receipts? For example: store details, receipt number, cashier, items, quantities, discounts, tax, payment method, and original sale reference.
- 20. May an authorized user reprint a receipt? If yes, should it be clearly marked as a reprint?
- 21. If receipt printing fails, should the completed sale remain successful and allow reprinting?

F. Inventory integration

- 22. Is the Inventory Service an existing external system accessed through an API?
- 23. Should inventory be updated only after payment succeeds?
- 24. If the Inventory Service is unavailable, should the POS:
 - Reject the sale
 - Complete the sale and queue the inventory update for retry
 - Allow a manager to decide

G. Returns and refunds

25. How is the original sale found: receipt number, barcode on the receipt, date, product, or customer/card information?
26. Are partial returns allowed, including returning only part of the quantity of an item?
27. What return rules apply? For example:
 - Return deadline
 - Original receipt required
 - Item must have been sold in that transaction
 - Quantity returned cannot exceed quantity purchased
 - A reason is mandatory
28. Can an item be returned more than once across separate return transactions, as long as the total returned quantity does not exceed the quantity sold?
29. Who may approve refunds: cashiers, managers, or cashiers only with manager authorization?
30. Must refunds always use the original payment method? For card refunds, must the refund go back through the Payment Service Provider to the original card transaction?
31. May a return be rejected after it has started? Can a completed refund ever be reversed?

H. Manager functions

32. For the daily sales summary, should the totals represent gross sales, refunds, cancellations, discounts, and net sales separately?
33. Does “selected date” use the store’s local calendar day, and can managers view past dates?
34. Should managers be able to print or export the daily summary?
35. What product information can managers maintain besides name and price, such as barcode, description, tax category, active status, and inventory identifier?
36. Should price changes take effect immediately? Must previous prices and the manager making each change be retained?

I. Non-functional expectations

37. Are there expected performance targets, such as adding an item or displaying the total within two seconds?
38. Must the POS continue operating during an internet or external-service outage?
39. Are there stated requirements for security, audit-log retention, backup, availability, supported hardware, or maximum number of checkout terminals?
40. Is this intended for one store only, or should the system support multiple stores and multiple POS terminals?

You are asking too many questions. Please assume you are a seasoned developer and when you see such requirements you kind of know based on your experience what it means. Where possible use best practices, so you ask fewer questions. Having said that, do not omit key questions that is clearly up to the stakeholder / user to answer and there's no good practice guidelines available. Please revise your list of clarification questions accordingly.

Agreed. I will apply standard POS practices for routine details such as barcode entry, quantities, cash change, unique receipt numbers, audit logs, receipt reprinting, and preventing duplicate card charges.

Only these stakeholder decisions need clarification:

1. **Tax:** Are product prices tax-inclusive or tax-exclusive? What tax rate or tax rules apply?
2. **Insufficient stock:** Should the POS block a sale when the Inventory Service reports insufficient stock, or warn the cashier and allow it?
3. **Discount policy:** What discounts are allowed, who can apply them, and can multiple discounts be combined?
4. **Payment methods:** Must a sale use either cash or card, or should split payment between cash and card be supported?
5. **External-service outage:** If the Payment Service Provider or Inventory Service is unavailable, must the POS stop the transaction, or should it support some form of offline or deferred processing?
6. **Return eligibility:** What is the return period, and is the original receipt mandatory? I will otherwise assume partial returns are permitted and total returned quantity cannot exceed the purchased quantity.
7. **Refund authorization:** Can cashiers process refunds independently, or does every refund require manager approval?
8. **Refund method:** Must a refund use the original payment method, or may store staff choose cash or card according to the situation?
9. **Deployment scope:** Is the system intended for one store and one POS terminal, one store with multiple terminals, or multiple stores?
10. **Daily summary:** Should the report show gross sales, discounts, refunds, net sales, and cash/card totals separately, or only the total cash and card amounts stated in the original requirements?

1. **Tax:** Are product prices tax-inclusive or tax-exclusive? What tax rate or tax rules apply?
Answer: Thai VAT Tax rules apply and each item is tax-inclusive.
1. **Insufficient stock:** Should the POS block a sale when the Inventory Service reports insufficient stock, or warn the cashier and allow it?
Answer: customer is bringing the item to the counter, so how this issue will never arise.
1. **Discount policy:** What discounts are allowed, who can apply them, and can multiple discounts be combined?
Answer: There is only one discount at the bottom.
1. **Payment methods:** Must a sale use either cash or card, or should split payment between cash and card be supported?
Answer: Do not allow split payment. Just cash or card.
1. **External-service outage:** If the Payment Service Provider or Inventory Service is unavailable, must the POS stop the transaction, or should it support some form of offline or deferred processing?
Answer: If PSP is not available then must pay by cash only.
1. **Return eligibility:** What is the return period, and is the original receipt mandatory? I will otherwise assume partial returns are permitted and total returned quantity cannot exceed the purchased quantity.
Answer: The POS system does not handle returns. That is handled at Customer Service Counter.
1. **Refund authorization:** Can cashiers process refunds independently, or does every refund require manager approval?
Answer: No refunds. Must got to customer service counter.

1. **Refund method:** Must a refund use the original payment method, or may store staff choose cash or card according to the situation?

Answer: No refunds via POS.

1. **Deployment scope:** Is the system intended for one store and one POS terminal, one store with multiple terminals, or multiple stores?

Answer: multiple terminals, multiple stores.

1. **Daily summary:** Should the report show gross sales, discounts, refunds, net sales, and cash/card totals separately, or only the total cash and card amounts stated in the original requirements?

Answer: Just total cash and card amounts.

Understood. I will treat returns, refunds, refund receipts, and returned-stock updates as outside the POS scope.

Only four important points remain:

1. **Discount:** Is the single order-level discount a percentage, a fixed amount, or either? Can cashiers enter it freely, or must it be selected from predefined discounts?
2. **Inventory outage:** If the Inventory Service is unavailable after a sale, should the POS complete the sale and retry the stock update later? This is my recommended approach.
3. **Customer Service integration:** Does the POS need to make completed sale information available to the Customer Service system, or is that integration outside this project?
4. **Daily summary scope:** Should a manager see the summary only for their store, or may selected managers view summaries for any store?

For physical stock discrepancies, I will assume the POS does not block the sale. It records the sale and reduces inventory, potentially exposing a stock discrepancy for later correction.

Discount: Is the single order-level discount a percentage, a fixed amount, or either? Can cashiers enter it freely, or must it be selected from predefined discounts?

Answer: There should be two textboxes; one for discount % and one for discount. If user enters/changes discount percent the discount amount is calculated, rounded to nearest satang - this amount is used as the discount amount. If user enters/changes the discount amount, the discount percent is set to null and this amount is used as the discount amount.

Inventory outage: If the Inventory Service is unavailable after a sale, should the POS complete the sale and retry the stock update later? This is my recommended approach.

Answer: Assume inventory service is not synchronous. A message is sent somewhere to record each sales transaction and gets processed by the inventory service system whenever it's available (asynchronously).

Customer Service integration: Does the POS need to make completed sale information available to the Customer Service system, or is that integration outside this project?

Answer: Outside

One answer appears to be missing:

Daily summary scope: Should a manager see the summary only for their store, or may selected managers view summaries for any store?

Daily summary scope: Should a manager see the summary only for their store, or may selected managers view summaries for any store?

Answer: Managers see their assigned store, while an administrator or head-office manager can select any store or view consolidated totals.

POS System Requirements, Draft 1

1. Functional Requirements

Checkout

FR-001 The system shall require each user to sign in under an authorized role, including Cashier, Manager, Administrator, or Head-Office Manager.

FR-002 The system shall allow an authorized cashier to start a new sale.

FR-003 The system shall allow the cashier to add products, change quantities, and remove products before the sale is completed.

FR-004 The system shall display and update the subtotal, discount, VAT information, and final amount whenever the contents of the sale change.

FR-005 The system shall provide one Discount Percentage field and one Discount Amount field for an order-level discount.

FR-006 The system shall allow the cashier to cancel a sale at any time before completion.

Payment

FR-007 The system shall allow the cashier to select either cash or credit card as the payment method.

FR-008 For a cash payment, the system shall accept the amount tendered and calculate the change due.

FR-009 For a credit-card payment, the system shall send the final sale amount to the connected card machine and Payment Service Provider.

FR-010 The system shall receive and display the approval or decline result returned by the Payment Service Provider.

FR-011 When a card payment is declined, the system shall allow the cashier to retry the card payment, change the payment method to cash, or cancel the sale.

FR-012 When the Payment Service Provider is unavailable, the system shall inform the cashier that card payment is unavailable and allow the cashier to accept cash or cancel the sale.

FR-013 The system shall record a completed sale with its sale identifier, store, terminal, cashier, date and time, sold items, discount, VAT, final amount, and payment method.

Receipt

FR-014 The system shall print a receipt using the store receipt printer after a sale is completed successfully.

Inventory Integration

FR-015 The system shall create an asynchronous inventory message for every completed sale.

FR-016 The inventory message shall contain a unique sale identifier, store identifier, terminal identifier, transaction date and time, and the product and quantity of each item sold.

FR-017 The system shall retain and retry an inventory message until it can be made available to the Inventory Service.

Sales Summary

FR-018 The system shall allow a manager to view the daily sales summary for the manager's assigned store and a selected date.

FR-019 The system shall allow an Administrator or Head-Office Manager to view the daily sales summary for any selected store.

FR-020 The system shall allow an Administrator or Head-Office Manager to view a consolidated daily sales summary across all stores.

FR-021 Each daily sales summary shall show the total amount received by cash and the total amount received by credit card separately.

Product and Price Management

FR-022 The system shall allow an authorized manager to add a product to the product catalog used by the POS.

FR-023 The system shall allow an authorized manager to update product information and selling prices within the manager's permitted scope.

2. Business Rules

BR-001 All product selling prices shall be expressed in Thai baht and shall include VAT according to applicable Thai VAT rules.

BR-002 The sale subtotal shall equal the sum of the extended amounts of all sale items before applying the order-level discount.

BR-003 A sale may have only one order-level discount. Product-line discounts and multiple combined discounts are not supported.

BR-004 When a user enters or changes the Discount Percentage, the system shall calculate the Discount Amount as:

$$\text{Discount Amount} = \text{Sale Subtotal} \times \text{Discount Percentage} \div 100$$

BR-005 A percentage-based Discount Amount shall be rounded to the nearest satang and shall be the amount used to calculate the final sale total.

BR-006 When a user enters or changes the Discount Amount directly, the system shall set the Discount Percentage to null and use the entered amount as the discount.

BR-007 When the sale contents change while a Discount Percentage is active, the system shall recalculate the Discount Amount using the revised subtotal.

BR-008 The Discount Percentage shall be between 0 and 100 percent, inclusive.

BR-009 The Discount Amount shall not be negative and shall not exceed the sale subtotal.

BR-010 The final sale amount shall equal the subtotal minus the Discount Amount.

BR-011 A sale shall use exactly one payment method. Split payment between cash and credit card is not permitted.

BR-012 A cash sale shall not be completed unless the amount tendered is equal to or greater than the final sale amount.

BR-013 A credit-card sale shall not be completed unless the Payment Service Provider approves payment for the final sale amount.

BR-014 A sale shall be considered completed after successful payment and permanent recording of the sale. Receipt-printing failure shall not reverse a completed sale.

BR-015 A cancelled sale shall not be included in sales totals and shall not produce an inventory message.

BR-016 The physical presence of an item at the checkout counter shall be treated as sufficient evidence that it may be sold. The POS shall not synchronously check or block a sale based on the recorded inventory balance.

BR-017 Daily cash and card totals shall include completed sales for the selected store or stores and selected local calendar date, after discounts.

BR-018 A Manager may view daily summaries only for the manager's assigned store.

BR-019 An Administrator or Head-Office Manager may view summaries for any individual store or for all stores combined.

BR-020 Returns, refunds, and refund receipts shall not be processed by the POS.

3. Non-Functional Requirements

NFR-001 Performance: Under normal operating conditions, the system shall respond to checkout actions, excluding external payment processing and physical printing, within two seconds for at least 95 percent of requests.

NFR-002 Reliability: A successfully recorded sale shall not be lost because the receipt printer, Payment Service Provider, messaging service, or Inventory Service becomes unavailable afterward.

NFR-003 Messaging Reliability: Inventory messages shall be stored durably and delivered using a mechanism that supports retry without losing completed sales.

NFR-004 Duplicate Prevention: Each inventory message shall contain a unique transaction identifier so that repeated delivery does not cause the Inventory Service to deduct the same stock more than once.

NFR-005 Payment Continuity: Failure of the Payment Service Provider shall not prevent the POS from completing an otherwise valid cash sale.

NFR-006 Security: The system shall enforce role-based access to checkout, reporting, and product-management functions.

NFR-007 Card Security: The POS shall not store sensitive card details and shall communicate with the card machine and Payment Service Provider through secure, approved interfaces.

NFR-008 Auditability: The system shall record the responsible user, date, time, store, and terminal for completed sales, sale cancellations, discounts, and product or price changes.

NFR-009 Scalability: The system shall support multiple stores and multiple concurrently operating POS terminals without transaction-identifier collisions.

NFR-010 Localization: The system shall represent monetary values in Thai baht to two decimal places and use the store's local date and time for daily reporting.

NFR-011 Interoperability: The system shall integrate with the approved receipt printer, card machine, Payment Service Provider, and asynchronous inventory-messaging interface.

4. Out of Scope

The following functions are outside the scope of this POS system:

1. Processing product returns
2. Issuing cash or card refunds
3. Printing refund receipts
4. Adding returned items back into inventory
5. Integration with the Customer Service Counter system
6. Split payments
7. Synchronous inventory availability checking

I am a bit concerned about this FR. What should happen if the inventory message cannot be written?

FR-017 The system shall retain and retry an inventory message until it can be made available to the Inventory Service.

That is a genuine weakness in FR-017. The system should distinguish:

- **Cannot store the message:** The sale must not be finalized because its inventory update could be permanently lost.
- **Cannot deliver the stored message:** The sale may be finalized because the durable message can be retried later.

The standard solution is a **transactional outbox**. The completed sale and its pending inventory-update record are written to the same database transaction. Either both are saved, or neither is saved.

I recommend replacing FR-017 with these requirements:

FR-017 The system shall durably record the completed sale and its pending inventory-update record as one atomic operation.

FR-018 If the system cannot record both the completed sale and its pending inventory-update record, it shall not mark the sale as completed and shall inform the cashier that the transaction could not be finalized.

FR-019 The system shall asynchronously send each pending inventory-update record to the inventory messaging service.

FR-020 If an inventory-update record cannot be delivered, the system shall retain it and retry delivery until the messaging service acknowledges receipt.

FR-021 Failure to deliver an already stored inventory-update record shall not reverse or block the corresponding completed sale.

FR-022 The system shall allow authorized personnel to view inventory-update records that are awaiting delivery or have repeatedly failed delivery.

This provides the important guarantee:

A completed sale can never exist without a durable pending inventory update.

There is one related card-payment edge case. If the card is approved but the POS fails while finalizing the sale, it must not charge the card again blindly. The payment should use a unique transaction reference so the POS can query or reconcile the existing payment result before retrying. This should be covered by an additional payment-recovery requirement.

In that case please rewrite the FR's for the card-payment edge case you mention. Then rewrite all the FR's for me so I can copy/paste over existing FR list (considering you have inserted a few FRs, so a renumbering is necessary).

Functional Requirements

Checkout

FR-001 The system shall require each user to sign in under an authorized role, including Cashier, Manager, Administrator, or Head-Office Manager.

FR-002 The system shall allow an authorized cashier to start a new sale.

FR-003 The system shall allow the cashier to add products, change quantities, and remove products before the sale is completed.

FR-004 The system shall display and update the subtotal, discount, VAT information, and final amount whenever the contents of the sale change.

FR-005 The system shall provide one Discount Percentage field and one Discount Amount field for an order-level discount.

FR-006 The system shall allow the cashier to cancel a sale at any time before completion.

Payment

FR-007 The system shall allow the cashier to select either cash or credit card as the payment method.

FR-008 For a cash payment, the system shall accept the amount tendered and calculate the change due.

FR-009 Before requesting card authorization, the system shall assign a unique reference to the card-payment attempt and record the pending payment attempt durably.

FR-010 If the system cannot record the pending card-payment attempt, it shall not send a payment request to the card machine or Payment Service Provider.

FR-011 For a credit-card payment, the system shall send the final sale amount and unique payment-attempt reference to the connected card machine and Payment Service Provider.

FR-012 The system shall receive and display the approval or decline result returned by the Payment Service Provider.

FR-013 When a card payment is declined, the system shall allow the cashier to retry the card payment, change the payment method to cash, or cancel the sale.

FR-014 When the Payment Service Provider is unavailable before a card-payment request is sent, the system shall inform the cashier that card payment is unavailable and allow the cashier to accept cash or cancel the sale.

FR-015 If communication is interrupted after a card-payment request has been sent, the system shall treat the payment result as unresolved and query the Payment Service Provider using the original payment-attempt reference.

FR-016 While a card-payment result remains unresolved, the system shall prevent the cashier from initiating another payment, changing to cash, or cancelling the sale.

FR-017 If card payment is approved but the system cannot finalize the sale, the system shall retain the payment attempt for recovery and shall not initiate another card charge.

FR-018 The system shall automatically reconcile unresolved card-payment attempts with the Payment Service Provider using their original payment-attempt references.

FR-019 If reconciliation confirms that a card payment was approved, the system shall finalize the sale without submitting another card charge.

FR-020 If reconciliation confirms that a card payment was declined or was not processed, the system shall return the sale to an unpaid state and allow the cashier to retry the card payment, change to cash, or cancel the sale.

FR-021 If the Payment Service Provider cannot determine the payment result, the system shall retain the payment attempt in a recovery-pending state and prevent another payment attempt until the result is resolved.

FR-022 The system shall allow authorized personnel to view card-payment attempts that remain unresolved or require recovery.

Sale Recording and Receipt

FR-023 The system shall record each completed sale with its sale identifier, store, terminal, cashier, date and time, sold items, discount, VAT, final amount, payment method, and applicable payment reference.

FR-024 The system shall print a receipt using the store receipt printer after a sale is completed successfully.

FR-025 If receipt printing fails, the system shall inform the cashier and allow an authorized user to reprint the receipt without reversing or duplicating the completed sale.

Inventory Integration

FR-026 When finalizing a sale, the system shall durably record the completed sale and its pending inventory-update record as one atomic operation.

FR-027 If the atomic operation cannot be completed, the system shall not mark the sale as completed. The system shall keep a cash sale open and inform the cashier, or place an already-approved card payment into the payment-recovery process.

FR-028 Each pending inventory-update record shall contain a unique sale identifier, store identifier, terminal identifier, transaction date and time, and the product and quantity of each item sold.

FR-029 The system shall asynchronously send each pending inventory-update record to the inventory messaging service.

FR-030 If an inventory-update record cannot be delivered, the system shall retain it and retry delivery until the inventory messaging service acknowledges receipt.

FR-031 Failure to deliver an already stored inventory-update record shall not reverse or block its corresponding completed sale.

FR-032 The system shall allow authorized personnel to view inventory-update records that are awaiting delivery or have repeatedly failed delivery.

Daily Sales Summary

FR-033 The system shall allow a Manager to view the daily sales summary for the Manager's assigned store and a selected date.

FR-034 The system shall allow an Administrator or Head-Office Manager to view the daily sales summary for any selected store.

FR-035 The system shall allow an Administrator or Head-Office Manager to view a consolidated daily sales summary across all stores.

FR-036 Each daily sales summary shall show the total amount received by cash and the total amount received by credit card separately.

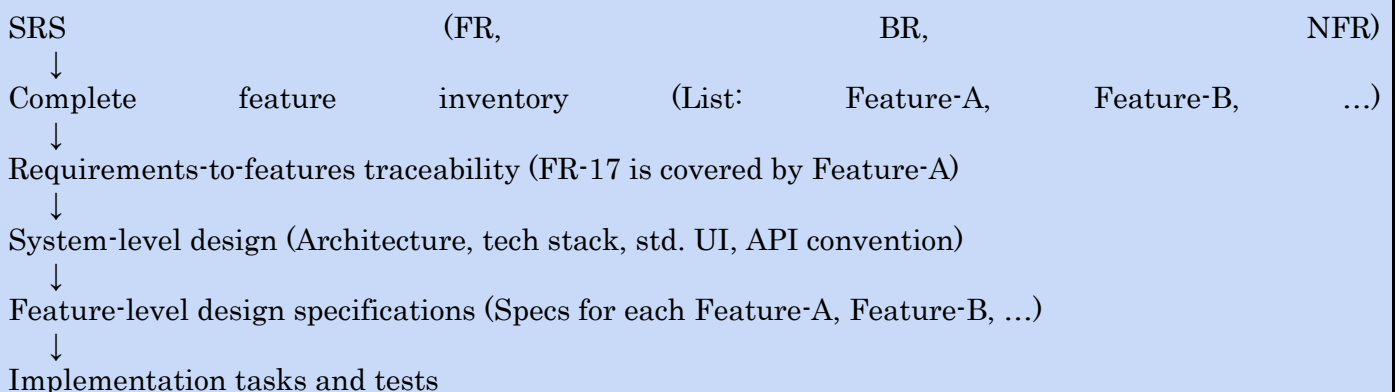
Product and Price Management

FR-037 The system shall allow an authorized Manager to add a product to the product catalog used by the POS.

FR-038 The system shall allow an authorized Manager to update product information and selling prices within the Manager's permitted scope.

Software Design Specification (SDS) Prompts

Once we have the SRS, the next step is to come up with the complete feature inventory list just to make sure we have covered all the requirements. After that we can design each feature. The suggested workflow is shown below. Can you list for me the Complete Feature Inventory for the POS app based on the FR, BR, NFR we have drafted above.



Complete Feature Inventory: POS System

Application Features

ID	Feature	Feature scope
Feature-A	User Authentication and Access Control	Sign-in, user roles, function permissions, and store-based access restrictions for Cashiers, Managers, Administrators, and Head-Office Managers.
Feature-B	Store and Terminal Operating Context	Identification of the current store and POS terminal, association of transactions with that context, and support for multiple stores and terminals.
Feature-C	Product Catalog and Price Management	Addition and modification of products, product information, VAT-inclusive selling prices, and manager access restrictions.
Feature-D	Sale Lifecycle Management	Starting a sale, maintaining its current state, cancelling it before completion, and preventing modification after completion.
Feature-E	Cart and Sale-Item Management	Adding products, changing quantities, removing products, and recalculating the sale when its items change.
Feature-F	VAT-Inclusive Pricing and Total Calculation	Calculation and display of item amounts, subtotal, Thai VAT information, discount, and final sale amount using Thai baht and satang.
Feature-G	Order-Level Discount Management	Discount Percentage and Discount Amount entry, percentage-to-amount calculation, satang rounding, manual discount-amount handling, validation, and recalculation when the cart changes.
Feature-H	Payment Method Selection	Selection of either cash or credit card, enforcement of the no-split-payment rule, and permitted switching of payment methods.
Feature-I	Cash Payment Processing	Entry of cash tendered, validation that sufficient cash was received, change calculation, and cash-sale completion.
Feature-J	Card Payment Authorization	Creation of a unique payment-attempt reference, durable recording of the attempt, communication with the card machine and Payment Service Provider, and handling of approved or declined results.
Feature-K	Card Payment Recovery and	Handling communication interruptions, unresolved payment results, approved payments that could not be

Can you show me the list of all requirements (NFR, FR, BR) and mention which feature each is covered under. This way I am certain all requirements have a feature to handle it.

This traceability matrix covers all 69 requirements. Every positive requirement maps to at least one feature. BR-020 is mapped to the explicit scope boundary because excluded functionality should not generate an implementation feature.

POS Requirements-to-Features Traceability Matrix

1. Functional Requirements

Requirement	Requirement statement	Covered by feature(s)
FR-001	The system shall require each user to sign in under an authorized role, including Cashier, Manager, Administrator, or Head-Office Manager.	Feature-A: User Authentication and Access Control
FR-002	The system shall allow an authorized cashier to start a new sale.	Feature-A: User Authentication and Access Control; Feature-D: Sale Lifecycle Management
FR-003	The system shall allow the cashier to add products, change quantities, and remove products before the sale is completed.	Feature-E: Cart and Sale-Item Management
FR-004	The system shall display and update the subtotal, discount, VAT information, and final amount whenever the contents of the sale change.	Feature-F: VAT-Inclusive Pricing and Total Calculation; Feature-G: Order-Level Discount Management
FR-005	The system shall provide one Discount Percentage field and one Discount Amount field for an order-level discount.	Feature-G: Order-Level Discount Management
FR-006	The system shall allow the cashier to cancel a sale at any time before completion.	Feature-D: Sale Lifecycle Management
FR-007	The system shall allow the cashier to select either cash or credit card as the payment method.	Feature-H: Payment Method Selection

FR-008	For a cash payment, the system shall accept the amount tendered and calculate the change due.	Feature-I: Cash Payment Processing
FR-009	Before requesting card authorization, the system shall assign a unique reference to the card-payment attempt and record the pending payment attempt durably.	Feature-J: Card Payment Authorization
FR-010	If the system cannot record the pending card-payment attempt, it shall not send a payment request to the card machine or Payment Service Provider.	Feature-J: Card Payment Authorization
FR-011	For a credit-card payment, the system shall send the final sale amount and unique payment-attempt reference to the connected card machine and Payment Service Provider.	Feature-J: Card Payment Authorization
FR-012	The system shall receive and display the approval or decline result returned by the Payment Service Provider.	Feature-J: Card Payment Authorization
FR-013	When a card payment is declined, the system shall allow the cashier to retry the card payment, change the payment method to cash, or cancel the sale.	Feature-J: Card Payment Authorization; Feature-H: Payment Method Selection; Feature-D: Sale Lifecycle Management
FR-014	When the Payment Service Provider is unavailable before a card-payment request is sent, the system shall inform the cashier that card payment is unavailable and allow the cashier to accept cash or cancel the sale.	Feature-J: Card Payment Authorization; Feature-H: Payment Method Selection
FR-015	If communication is interrupted after a card-payment request has been sent, the system shall treat the payment result as unresolved and query the Payment Service Provider using the original payment-attempt reference.	Feature-K: Card Payment Recovery and Reconciliation

FR-016	While a card-payment result remains unresolved, the system shall prevent the cashier from initiating another payment, changing to cash, or cancelling the sale.	Feature-K: Card Payment Recovery and Reconciliation
FR-017	If card payment is approved but the system cannot finalize the sale, the system shall retain the payment attempt for recovery and shall not initiate another card charge.	Feature-K: Card Payment Recovery and Reconciliation; Feature-L: Sale Finalization and Durable Recording
FR-018	The system shall automatically reconcile unresolved card-payment attempts with the Payment Service Provider using their original payment-attempt references.	Feature-K: Card Payment Recovery and Reconciliation
FR-019	If reconciliation confirms that a card payment was approved, the system shall finalize the sale without submitting another card charge.	Feature-K: Card Payment Recovery and Reconciliation; Feature-L: Sale Finalization and Durable Recording
FR-020	If reconciliation confirms that a card payment was declined or was not processed, the system shall return the sale to an unpaid state and allow the cashier to retry the card payment, change to cash, or cancel the sale.	Feature-K: Card Payment Recovery and Reconciliation; Feature-H: Payment Method Selection; Feature-D: Sale Lifecycle Management
FR-021	If the Payment Service Provider cannot determine the payment result, the system shall retain the payment attempt in a recovery-pending state and prevent another payment attempt until the result is resolved.	Feature-K: Card Payment Recovery and Reconciliation
FR-022	The system shall allow authorized personnel to view card-payment attempts that remain unresolved or require recovery.	Feature-P: Operational Exception Monitoring; Feature-A: User Authentication and Access Control
FR-023	The system shall record each completed sale with its sale identifier, store, terminal, cashier, date and time, sold items, discount, VAT, final amount, payment method, and applicable payment reference.	Feature-L: Sale Finalization and Durable Recording; Feature-B: Store and Terminal Operating Context

FR-024	The system shall print a receipt using the store receipt printer after a sale is completed successfully.	Feature-M: Receipt Printing and Reprinting
FR-025	If receipt printing fails, the system shall inform the cashier and allow an authorized user to reprint the receipt without reversing or duplicating the completed sale.	Feature-M: Receipt Printing and Reprinting
FR-026	When finalizing a sale, the system shall durably record the completed sale and its pending inventory-update record as one atomic operation.	Feature-L: Sale Finalization and Durable Recording; Feature-N: Inventory Update Outbox
FR-027	If the atomic operation cannot be completed, the system shall not mark the sale as completed. The system shall keep a cash sale open and inform the cashier, or place an already-approved card payment into the payment-recovery process.	Feature-L: Sale Finalization and Durable Recording; Feature-I: Cash Payment Processing; Feature-K: Card Payment Recovery and Reconciliation; Feature-N: Inventory Update Outbox
FR-028	Each pending inventory-update record shall contain a unique sale identifier, store identifier, terminal identifier, transaction date and time, and the product and quantity of each item sold.	Feature-N: Inventory Update Outbox; Feature-B: Store and Terminal Operating Context
FR-029	The system shall asynchronously send each pending inventory-update record to the inventory messaging service.	Feature-O: Inventory Message Delivery
FR-030	If an inventory-update record cannot be delivered, the system shall retain it and retry delivery until the inventory messaging service acknowledges receipt.	Feature-O: Inventory Message Delivery
FR-031	Failure to deliver an already stored inventory-update record shall not reverse or block its corresponding completed sale.	Feature-O: Inventory Message Delivery; Feature-L: Sale Finalization and Durable Recording

FR-032	The system shall allow authorized personnel to view inventory-update records that are awaiting delivery or have repeatedly failed delivery.	Feature-P: Operational Exception Monitoring; Feature-O: Inventory Message Delivery
FR-033	The system shall allow a Manager to view the daily sales summary for the Manager's assigned store and a selected date.	Feature-Q: Daily Sales Reporting; Feature-A: User Authentication and Access Control
FR-034	The system shall allow an Administrator or Head-Office Manager to view the daily sales summary for any selected store.	Feature-Q: Daily Sales Reporting; Feature-A: User Authentication and Access Control
FR-035	The system shall allow an Administrator or Head-Office Manager to view a consolidated daily sales summary across all stores.	Feature-Q: Daily Sales Reporting; Feature-A: User Authentication and Access Control
FR-036	Each daily sales summary shall show the total amount received by cash and the total amount received by credit card separately.	Feature-Q: Daily Sales Reporting
FR-037	The system shall allow an authorized Manager to add a product to the product catalog used by the POS.	Feature-C: Product Catalog and Price Management; Feature-A: User Authentication and Access Control
FR-038	The system shall allow an authorized Manager to update product information and selling prices within the Manager's permitted scope.	Feature-C: Product Catalog and Price Management; Feature-A: User Authentication and Access Control

2. Business Rules

Requirement	Requirement statement	Covered by feature(s)
BR-001	All product selling prices shall be expressed in Thai baht and shall include VAT according to applicable Thai VAT rules.	Feature-C: Product Catalog and Price Management; Feature-F: VAT-Inclusive Pricing and Total Calculation

BR-002	The sale subtotal shall equal the sum of the extended amounts of all sale items before applying the order-level discount.	Feature-E: Cart and Sale-Item Management; Feature-F: VAT-Inclusive Pricing and Total Calculation
BR-003	A sale may have only one order-level discount. Product-line discounts and multiple combined discounts are not supported.	Feature-G: Order-Level Discount Management
BR-004	When a user enters or changes the Discount Percentage, the system shall calculate the Discount Amount as: Discount Amount = Sale Subtotal × Discount Percentage ÷ 100.	Feature-G: Order-Level Discount Management
BR-005	A percentage-based Discount Amount shall be rounded to the nearest satang and shall be the amount used to calculate the final sale total.	Feature-G: Order-Level Discount Management; Feature-F: VAT-Inclusive Pricing and Total Calculation
BR-006	When a user enters or changes the Discount Amount directly, the system shall set the Discount Percentage to null and use the entered amount as the discount.	Feature-G: Order-Level Discount Management
BR-007	When the sale contents change while a Discount Percentage is active, the system shall recalculate the Discount Amount using the revised subtotal.	Feature-G: Order-Level Discount Management; Feature-E: Cart and Sale-Item Management
BR-008	The Discount Percentage shall be between 0 and 100 percent, inclusive.	Feature-G: Order-Level Discount Management
BR-009	The Discount Amount shall not be negative and shall not exceed the sale subtotal.	Feature-G: Order-Level Discount Management
BR-010	The final sale amount shall equal the subtotal minus the Discount Amount.	Feature-F: VAT-Inclusive Pricing and Total Calculation; Feature-G: Order-Level Discount Management
BR-011	A sale shall use exactly one payment method. Split payment between cash and credit card is not permitted.	Feature-H: Payment Method Selection

BR-012	A cash sale shall not be completed unless the amount tendered is equal to or greater than the final sale amount.	Feature-I: Cash Payment Processing
BR-013	A credit-card sale shall not be completed unless the Payment Service Provider approves payment for the final sale amount.	Feature-J: Card Payment Authorization; Feature-L: Sale Finalization and Durable Recording
BR-014	A sale shall be considered completed after successful payment and permanent recording of the sale. Receipt-printing failure shall not reverse a completed sale.	Feature-L: Sale Finalization and Durable Recording; Feature-M: Receipt Printing and Reprinting
BR-015	A cancelled sale shall not be included in sales totals and shall not produce an inventory message.	Feature-D: Sale Lifecycle Management; Feature-Q: Daily Sales Reporting; Feature-N: Inventory Update Outbox
BR-016	The physical presence of an item at the checkout counter shall be treated as sufficient evidence that it may be sold. The POS shall not synchronously check or block a sale based on the recorded inventory balance.	Feature-E: Cart and Sale-Item Management; Feature-N: Inventory Update Outbox
BR-017	Daily cash and card totals shall include completed sales for the selected store or stores and selected local calendar date, after discounts.	Feature-Q: Daily Sales Reporting
BR-018	A Manager may view daily summaries only for the Manager's assigned store.	Feature-A: User Authentication and Access Control; Feature-Q: Daily Sales Reporting
BR-019	An Administrator or Head-Office Manager may view summaries for any individual store or for all stores combined.	Feature-A: User Authentication and Access Control; Feature-Q: Daily Sales Reporting
BR-020	Returns, refunds, and refund receipts shall not be processed by the POS.	Scope boundary: Verified by the absence of return, refund, and refund-receipt features. No implementation feature is required.

3. Non-Functional Requirements

Requirement	Requirement statement	Covered by feature(s)
NFR-001	Under normal operating conditions, the system shall respond to checkout actions, excluding external payment processing and physical printing, within two seconds for at least 95 percent of requests.	Feature-D: Sale Lifecycle Management; Feature-E: Cart and Sale-Item Management; Feature-F: VAT-Inclusive Pricing and Total Calculation; Feature-G: Order-Level Discount Management; Feature-H: Payment Method Selection; Feature-I: Cash Payment Processing; Feature-L: Sale Finalization and Durable Recording
NFR-002	A successfully recorded sale shall not be lost because the receipt printer, Payment Service Provider, messaging service, or Inventory Service becomes unavailable afterward.	Feature-L: Sale Finalization and Durable Recording; Feature-M: Receipt Printing and Reprinting; Feature-J: Card Payment Authorization; Feature-K: Card Payment Recovery and Reconciliation; Feature-N: Inventory Update Outbox; Feature-O: Inventory Message Delivery
NFR-003	Inventory messages shall be stored durably and delivered using a mechanism that supports retry without losing completed sales.	Feature-N: Inventory Update Outbox; Feature-O: Inventory Message Delivery
NFR-004	Each inventory message shall contain a unique transaction identifier so that repeated delivery does not cause the Inventory Service to deduct the same stock more than once.	Feature-N: Inventory Update Outbox; Feature-O: Inventory Message Delivery
NFR-005	Failure of the Payment Service Provider shall not prevent the POS from completing an otherwise valid cash sale.	Feature-H: Payment Method Selection; Feature-I: Cash Payment Processing; Feature-J: Card Payment Authorization
NFR-006	The system shall enforce role-based access to checkout, reporting, and product-management functions.	Feature-A: User Authentication and Access Control

NFR-007	The POS shall not store sensitive card details and shall communicate with the card machine and Payment Service Provider through secure, approved interfaces.	Feature-J: Card Payment Authorization; Feature-K: Card Payment Recovery and Reconciliation
NFR-008	The system shall record the responsible user, date, time, store, and terminal for completed sales, sale cancellations, discounts, and product or price changes.	Feature-R: Audit Trail
NFR-009	The system shall support multiple stores and multiple concurrently operating POS terminals without transaction-identifier collisions.	Feature-B: Store and Terminal Operating Context; Feature-L: Sale Finalization and Durable Recording
NFR-010	The system shall represent monetary values in Thai baht to two decimal places and use the store's local date and time for daily reporting.	Feature-F: VAT-Inclusive Pricing and Total Calculation; Feature-B: Store and Terminal Operating Context; Feature-Q: Daily Sales Reporting
NFR-011	The system shall integrate with the approved receipt printer, card machine, Payment Service Provider, and asynchronous inventory-messaging interface.	Feature-M: Receipt Printing and Reprinting; Feature-J: Card Payment Authorization; Feature-O: Inventory Message Delivery

4. Coverage Summary

Requirement type	Total	Mapped to features	Scope-boundary requirements	Unmapped
Functional Requirements	38	38	0	0
Business Rules	20	19	1	0
Non-Functional Requirements	11	11	0	0

Total	69	68	1	0
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All requirements have traceability coverage. There are no unmapped requirements.

Next I should work on the system level SDS for my POS App.

Here is some information you may need:

- Tech Stack: 1) React + TypeScript + Vite 2) Node.js + Express + TypeScript 3) PostgreSQL via Prisma ORM
- Product Name: "Kaching" For theme color:
- Use KMUTT university color palette.

I'm sure you are missing a lot of information before you can draft the system level SDS. Please ask me for each first while also providing me a recommendation. After answering you can provide me the a sample system level SDS for me to edit.

I will assume standard practices for REST APIs, validation, error responses, database migrations, logging, TypeScript conventions, automated testing, and secure configuration. The following decisions materially affect the architecture or reveal possible SRS gaps.

You can reply “accept all recommendations” and list only exceptions.

1. Hosting and deployment

Recommendation: Use a centralized cloud architecture:

- React application served to browser-based POS terminals
- Express API deployed as Docker containers
- Central PostgreSQL database
- Separate background worker for asynchronous processing
- Development with Docker Compose
- Production on Google Cloud using Cloud Run and Cloud SQL

Do you accept this deployment model?

2. Store internet outage

The SRS explains what happens when the PSP or Inventory Service is unavailable, but not when a terminal loses access to the Kaching backend.

Recommendation: Kaching requires connectivity to its central backend. If connectivity is lost, the terminal cannot start or complete new sales. Full offline sales would require a local database, synchronization, and conflict-handling architecture.

Should Kaching support offline sales, or require backend connectivity?

3. External-system implementations

For a student project, real banking and hardware integrations may not be available.

Recommendation:

- Implement a simulated K-Bank card machine and PSP
- Define a PSP adapter interface so a real provider can replace the simulator
- Simulate the Inventory Service consumer
- Support browser printing initially, behind a receipt-printer adapter

Should the SDS describe simulated integrations or actual production integrations?

4. Asynchronous inventory messaging

Recommendation: Use the PostgreSQL transactional outbox for durable recording and Google Cloud Pub/Sub for message delivery. A background worker reads pending outbox records, publishes them, and records acknowledgements.

If you do not want GCP-specific messaging, RabbitMQ is a suitable alternative.

Which should the SDS use: Google Cloud Pub/Sub or RabbitMQ?

5. User, store, and terminal administration

The current requirements use accounts, roles, store assignments, and terminals, but do not explain how they are created or maintained.

Recommendation: Add an Administration feature allowing Administrators to:

- Create and deactivate users
- Assign roles and stores
- Create and update stores
- Register and deactivate POS terminals

Should this feature be added to the SRS and feature inventory?

6. Product and price scope

It is not yet clear whether products and prices are shared across stores.

Recommendation: Use one global product catalog and one global selling price per product for the first version. Authorized Managers or Head-Office Managers maintain them, and changes apply to all stores.

Do you accept global products and prices, or should each store have its own prices?

7. Product entry at checkout

Recommendation: Support:

- Barcode scanning through a keyboard-style USB barcode scanner
- Manual barcode entry
- Product search by code or name

Should all three methods be included?

8. Capacity and recovery targets

Recommendation: Design the sample system for:

- Up to 20 stores
- Up to 10 active terminals per store
- Up to 200 concurrent users
- At least 20 completed sales per second across the organization
- 99.5% monthly backend availability
- Recovery Point Objective of 15 minutes
- Recovery Time Objective of 4 hours

Are these reasonable targets for your project?

9. User interface and language

Recommendation:

- English user interface for the first version
- Thai baht, Thai VAT, and Asia/Bangkok business time
- Desktop and touchscreen-friendly layout
- Minimum supported resolution of 1280×720
- KMUTT orange as the primary brand color, supported by white, dark neutral, and accessible status colors
- WCAG 2.1 AA color contrast

Should the application be English-only, Thai-only, or bilingual?

1. Hosting and deployment

Recommendation: Use a centralized cloud architecture:

React application served to browser-based POS terminals

Express API deployed as Docker containers

Central PostgreSQL database

Separate background worker for asynchronous processing

Development with Docker Compose

Production on Google Cloud using Cloud Run and Cloud SQL

Do you accept this deployment model? YES

2. Store internet outage

The SRS explains what happens when the PSP or Inventory Service is unavailable, but not when a terminal loses access to the Kaching backend.

Recommendation: Kaching requires connectivity to its central backend. If connectivity is lost, the terminal cannot start or complete new sales. Full offline sales would require a local database, synchronization, and conflict-handling architecture.

Should Kaching support offline sales, or require backend connectivity? No, let's keep it simple.

3. External-system implementations

For a student project, real banking and hardware integrations may not be available.

Recommendation:

Implement a simulated K-Bank card machine and PSP

Define a PSP adapter interface so a real provider can replace the simulator

Simulate the Inventory Service consumer

Support browser printing initially, behind a receipt-printer adapter

Should the SDS describe simulated integrations or actual production integrations? Use actual production integration - we will not implement this but want a good example SDS.

4. Asynchronous inventory messaging

Recommendation: Use the PostgreSQL transactional outbox for durable recording and Google Cloud Pub/Sub for message delivery. A background worker reads pending outbox records, publishes them, and records acknowledgements.

If you do not want GCP-specific messaging, RabbitMQ is a suitable alternative.

Which should the SDS use: Google Cloud Pub/Sub or RabbitMQ? RabbitMQ

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Create and deactivate users

Assign roles and stores

Create and update stores

Register and deactivate POS terminals

Should this feature be added to the SRS and feature inventory? Yes

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It is not yet clear whether products and prices are shared across stores.

Recommendation: Use one global product catalog and one global selling price per product for the first version. Authorized Managers or Head-Office Managers maintain them, and changes apply to all stores.

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Manual barcode entry

Product search by code or name

Should all three methods be included? Yes

8. Capacity and recovery targets

Recommendation: Design the sample system for:

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Up to 10 active terminals per store

Up to 200 concurrent users

At least 20 completed sales per second across the organization

99.5% monthly backend availability

Recovery Point Objective of 15 minutes

Recovery Time Objective of 4 hours

Are these reasonable targets for your project? Yes

9. User interface and language

Recommendation:

English user interface for the first version

Thai baht, Thai VAT, and Asia/Bangkok business time

Desktop and touchscreen-friendly layout

Minimum supported resolution of 1280 × 720

KMUTT orange as the primary brand color, supported by white, dark neutral, and accessible status colors

WCAG 2.1 AA color contrast

Should the application be English-only, Thai-only, or bilingual? English-only